Carry: One-Tap Delta-Neutral Dollar Yield on Monad

Prepared for Hooyar Lotfy | Monad Metropolis Hackathon | Track 01: Onchain Finance & Trading | Draft of October 10, 2026

1. Executive Summary

Carry is a mobile app that lets someone who does not think of themselves as a trader earn market-neutral yield on a dollar balance. With one tap it buys a token on Kuru (spot order book) and opens an equal-sized short on Perpl (perpetuals) on Monad. The price exposure of the two legs cancels. What remains is the funding payment that perp longs pay shorts when the market is bullish. The user sees a dollar balance (AUSD), a live net APR estimate, a hedge-health meter, and an Unwind button. They never see a seed phrase: login is a passkey through Mera.

The project is built to stack sponsor bounties instead of chasing one: Agora (mobile trading app, Mera auth, AUSD balance, trades via Perpl), Kuru (consumer spot product on its order book), Perpl (production-grade API automation), Monad Foundation (Mera-powered UX and creative PRF use) and Envio (indexed data driving a core feature). Its Monad story is concrete: two onchain venues, one decision engine, sub-second execution on both legs, and a rebalancer that stays cheap enough to run continuously.

Honesty note that shapes the whole design: carry is not risk-free yield. Funding can turn negative, a leg can fail to fill, and a violent price move can stress the margin on the perp leg. The product is built around making those risks visible and bounded, not hiding them. The central defense is the Risk Guard (section 5.5): a three-tier margin ladder, an AUSD reserve buffer and an emergency unwind that keep the short leg far from liquidation. Nothing in this document is financial advice or a forecast.

2. Problem and Target User

Problem. Funding-rate carry is one of the most discussed market-neutral strategies in crypto, yet it is operationally ugly for normal people: two venues, two accounts, two order tickets, manual sizing, manual margin management, and constant monitoring. Most retail

users who try it get the hedge ratio wrong or miss a margin top-up. Meanwhile perp open interest on decentralized venues is at record levels and retail trading demand (copy trading, one-login trading apps) is rising, so there is demand for packaged, safer ways in.

Primary persona: Selin, 29, Istanbul. Holds dollars as a hedge against local currency risk. Understands 'earn yield on dollars' but not funding rates. Wants to see a balance, a yield number and a way out. Will not install a browser extension or write down 12 words.

Secondary persona: Diego, 34, Buenos Aires. Already uses a stablecoin wallet (Lemon-style). Wants to move from passive holding to yield without learning perps.

Judge-facing framing. The judges are protocol founders and investors. The pitch is: we turned a two-venue pro strategy into a consumer primitive, and only a chain with Monad's speed and low per-action cost makes continuous rebalancing from a phone economical.

3. Product Concept

3.1 How the strategy works

1. The user deposits AUSD (or already holds it).
2. The app chooses an eligible market and a leverage setting for the perp leg (default 1x with a 10% AUSD reserve buffer, hard cap 2x).
3. It sizes both legs to the same base quantity. Example with 1,000 AUSD of capital, 1x leverage and a 10% reserve: 100 AUSD stays in reserve, about 450 AUSD buys spot on Kuru, and about 450 AUSD is margin for a 450 AUSD short on Perpl.
4. Spot gains offset perp losses and vice versa, so net price exposure is near zero.
5. While funding is positive, shorts are paid. The user earns that stream minus fees and slippage.
6. A monitor watches hedge drift, margin health and funding. It rebalances or exits automatically within the user's risk settings.

Illustrative math (not a forecast). If funding annualizes to 15% on notional, yield on total capital is roughly 15% x L/(L+1). At the 1x default with a 10% reserve it is about 6.75% gross (7.5% without the reserve, 10% at the 2x opt-in), because safety is paid for with idle margin and cash. If round-trip costs (fees plus slippage) are 0.20% of notional, daily carry at 15% APR is about 0.041% of notional, so break-even is roughly 5 days. The app should display this break-even before the user confirms.

3.2 What the user sees

Screen	Purpose	Key elements
Welcome	Zero-friction onboarding	'Unlock Vault' with Face ID, no wallet language
Home	Balance and yield	AUSD balance, live net APR estimate, net delta in dollars (green when near zero), two leg cards (Kuru spot long, Perpl short)
Start Earning	One-tap entry	Amount slider, market chosen by engine, break-even days, risk meter, confirm
Position	Trust and control	Hedge ratio, drift, funding accrued, margin ladder with tier lines, distance to liquidation, guard incident log, Unwind
Activity	Proof	Every leg with transaction links, powered by Envio
Spot	Kuru-focused view	Order book depth and a plain spot buy, so the Kuru bounty sees a real spot product
Settings	Risk policy	Max leverage (1x default, 2x advanced), reserve buffer size, max slippage, auto-exit funding floor, alert preferences

3.3 Principles

- **Dollars in, dollars out.** The user's unit of account is AUSD.
- **Show the risk, bounded.** Every position displays its liquidation distance and the conditions under which the app will auto-exit.
- **Reversible.** Unwind is always one tap and always works, even if funding data is stale.
- **Proof over promises.** Every hedge action is anchored by an onchain record that anyone can verify.

4. Why Monad

The scoring rubric gives Monad Integration 20%, and generic EVM ports are explicitly penalized. Carry's dependence on Monad is structural:

- **Two venues, one clock.** Both the spot order book and the perp venue live on Monad, so both legs can be sent and confirmed within roughly a second. On a slower chain, the window where only one leg exists (naked price exposure) is much longer.
- **Continuous rebalancing is economical.** Drift correction and margin top-ups are small, frequent actions. Low fees and fast blocks make this viable from a phone.
- **Order-book native.** Kuru is a fully onchain order book, which suits limit-order entries that reduce slippage on the spot leg.
- **Engineering detail worth demonstrating.** Monad charges the gas limit you declare, not gas used. The app sets tight per-call gas limits, and the README should document this as a Monad-specific optimization.
- **Event-rich chain, so indexing matters.** Envio HyperIndex gives live position history without a bespoke indexer.

5. Architecture

5.1 Components

```
[Mobile app: Expo / React Native]
   | Mera passkey -> viem account (no seed phrase)
   | Strategy engine (TypeScript, runs in-app)
   | Risk Guard: margin ladder, AUSD reserve buffer, circuit breakers
   |        |-- Kuru: spot leg (onchain order book, SDK or direct contract
calls)
   |        |-- Perpl: perp leg (API with trade-only key), mark price,
backstop stops
   |        '-- CarryLedger contract: hedge state, risk policy, guard events
   '-- Envio HyperIndex GraphQL: positions, funding, drift, guard history
[Optional Phase 2: remote keeper using a time-boxed trade-only key]
```

Layer	Choice	Reason
Mobile	Expo (React Native)	Agora bounty requires a mobile app; Mera ships React Native support (v0.2.0)
Auth	Mera + react-native-passkey	Passkey-derived Monad account; same account on web and mobile if rpId matches

Layer	Choice	Reason
Chain access	viem, chain 143 (mainnet) / 10143 (testnet)	Standard; Mera exposes a viem account adapter
Spot venue	Kuru onchain order book	Kuru bounty; published market-maker SDKs show event and order APIs
Perp venue	Perpl API	Perpl bounty; trade-only keys that cannot withdraw are supported
Risk layer	Risk Guard (TypeScript, in-app)	Keeps the short far from liquidation: tiered margin ladder, AUSD reserve buffer, circuit breakers (section 5.5)
Indexer	Envio HyperIndex	Envio bounty; powers Activity, PnL and guard history
Own contract	CarryLedger (Solidity, Foundry)	Satisfies 'verified contract addresses', adds auditable proof
Hosting	Static docs site plus public repo	README must be reproducible

5.2 The CarryLedger contract

The contract is deliberately small and non-custodial. It never holds user funds.

- `openHedge(marketId, baseQty, spotRef, perpRef, leverage)` records a position owned by msg.sender and emits `HedgeOpened`.
- `rebalance(positionId, newSpotQty, newPerpQty, reason)` emits `HedgeRebalanced` with before/after hedge ratio.
- `closeHedge(positionId, realizedPnl, reason)` emits `HedgeClosed`.
- `setPolicy(maxLeverage, maxSlippageBps, fundingFloorBps, maxDriftBps)` stores the user's risk policy; the app reads it back so the policy is portable across devices. It also holds the Risk Guard settings (tier thresholds, reserve buffer size, maximum leverage), and the ledger rejects entries that exceed the policy's leverage cap.
- Guards: only the owner can mutate a position; hedge ratio sanity bounds; reentrancy not applicable (no value transfer), but use custom errors and tight gas limits.
- Events are the product: Envio indexes them to render 'proof of hedge' for each position, including the GuardTriggered and EmergencyUnwind records written by the Risk Guard.

Why a ledger and not a vault: a vault would add custody risk, audit burden and scope that cannot be finished safely in three days. A ledger gives judges clean contract logic and a verifiable trail with minimal attack surface. A pooled vault is a roadmap item.

5.3 Where the perp leg lives

Perpl is accessed through its API with a trade-only key. Evidence from a third-party integration guide: the user enrolls a key with an EIP-712 signature, the key can trade but never withdraw, and it can be revoked. That matters for security design and for the Mera bounty (see 5.4). **Assumption to verify on day one:** which Perpl actions are onchain versus API-matched, whether AUSD is accepted as collateral, the exact maintenance margin requirement and liquidation penalty, and whether reduce-only stop orders exist (this decides the exchange-resident backstop in section 5.5). A DeFi dashboard lists Perpl holding AUSD on Monad, which is encouraging but not proof.

5.4 Keeping the Mera 'no custody backend' promise

The 'Best Mera-Powered UX' bounty requires Mera to be the entire account layer with no seed phrase, no extension and no custody backend. A keeper that holds API keys would violate the spirit of that. Design:

- **Core build (bounty-safe):** the strategy engine runs inside the app. The Perpl trade-only key is encrypted with a key derived from the passkey's PRF output (Mera supports using PRF to encrypt an independent secret, not only to derive wallet keys). The ciphertext may sit in device storage or the CarryLedger metadata; only the passkey can decrypt it. The app decrypts into memory per session and clears buffers afterward.
- **Trade-off, stated openly:** auto-rebalance works while the app is active or backgrounded within OS limits. For always-on protection, Phase 2 offers an opt-in remote keeper that receives the key for a time-boxed session. Ship this only if the bounty text allows it and mark it clearly in the README.
- **Never** store PRF output in AsyncStorage or logs; use biometric-protected secure storage.

PRF vault reference implementation (web path). The Perpl trade-only key is encrypted under a key derived from the passkey's PRF output. Use Mera's helper rather than raw `navigator.credentials`, derive with HKDF instead of importing raw bytes as a key, and use a fresh IV for every encryption:

```
import { getPasskeyPrfOutput } from '@category-labs/mera';

async function vaultKey(rpId: string): Promise<CryptoKey> {
  const { prfOutput } = await getPasskeyPrfOutput({ rpId });
  const ikm = await crypto.subtle.importKey('raw', prfOutput, 'HKDF', false,
['deriveKey']);
  return crypto.subtle.deriveKey(
    { name: 'HKDF', hash: 'SHA-256', salt: new Uint8Array(32), info: new
TextEncoder().encode('carry/perpl-api-key/v1') },
    ikm, { name: 'AES-GCM', length: 256 }, false, ['encrypt', 'decrypt']);
}
// Encrypt with a random 12-byte IV and store iv + ciphertext.
// Never log or persist prfOutput or the derived key.
```

- Verify the helper's call shape against the installed Mera version; the snippet follows Mera's published example.
- `navigator.credentials` does not exist in React Native. On the Expo app use Mera's React Native WebAuthn client with react-native-passkey, plus a WebCrypto-compatible library for HKDF and AES-GCM.
- Encrypt the real Perpl trade-only key, not a dummy. The ciphertext may sit in device storage; the derived key never does.
- On desktop Chrome only passkeys saved to Google Password Manager return PRF, so test on the target phone first.

5.5 Anti-Risk System: The Risk Guard

A liquidated short leg turns a hedge into naked spot, so the Risk Guard keeps margin health far above the liquidation line using three escalating tiers, an AUSD reserve buffer and an exchange-resident backstop.

Why this is the central risk. Take 1,000 AUSD: 500 buys spot on Kuru and 500 is margin for a 1x short (500 notional) on Perpl. Total value stays near 1,000 while hedged, but the short is liquidated if price rises far enough. The table assumes a maintenance margin requirement (MMR) of 5% of notional; verify Perpl's actual figure.

Token price change	Kuru spot value	Perpl short PnL	Perpl margin left	Margin ratio	Net value	Status
0%	$500	$0	$500	100%	$1,000	Safe
+50%	$750	-$250	$250	33%	$1,000	Safe

Token price change	Kuru spot value	Perpl short PnL	Perpl margin left	Margin ratio	Net value	Status
+80%	$900	-$400	$100	11%	$1,000	Warning zone
+92%	$960	-$460	$40	4%	about $960	Liquidated (below MMR)

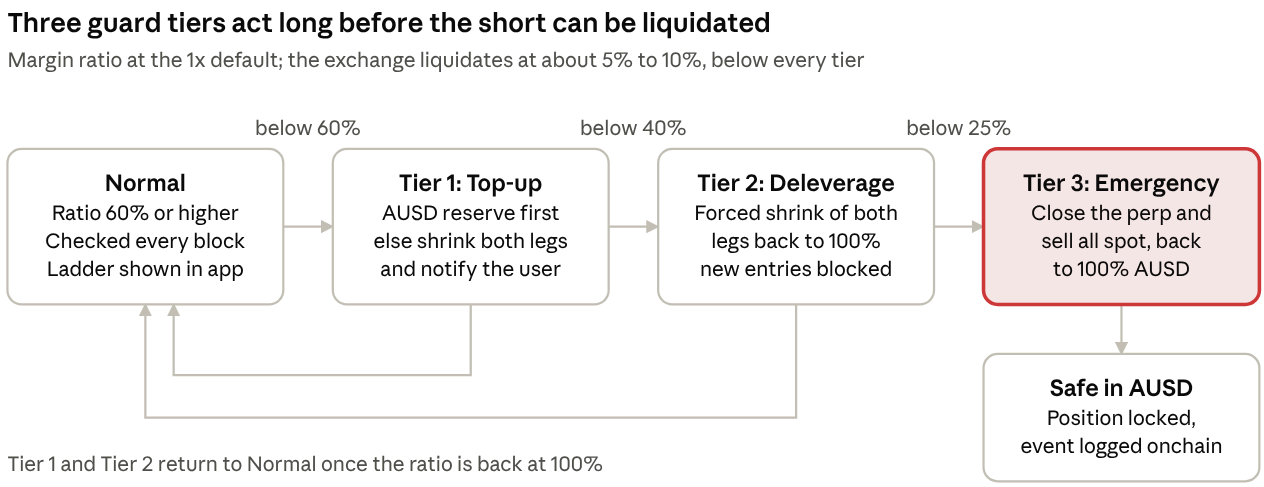
Margin ratio = perp margin left divided by the current short notional. Liquidation does not destroy value by itself; it destroys the hedge. Afterwards the user holds naked spot, loses part of the remaining margin to the liquidation penalty, and any price drop becomes a direct loss of principal. The Risk Guard exists to make that state unreachable.

Rebalancing must shrink both legs. Topping up margin by selling spot only works if the same base quantity of the short is closed too; otherwise spot quantity falls below short quantity and the position turns net short. The guard therefore recycles margin in this order: (1) draw from the AUSD reserve buffer; (2) if it is empty, sell a fraction f of spot, close the same fraction f of the short, and add the proceeds to margin. Worked example at +30% with 1x: spot is worth 650, margin left is 350, short notional is 650. Setting f = 30% shrinks spot and short to 455 each and lifts margin to 545, a ratio of about 120%, still delta-neutral and still worth 1,000 in total.

The three-tier ladder (margin ratio thresholds for the 1x default):

Tier	Trigger (margin ratio)	Approx. price rise at 1x	Action
Normal	60% or higher	up to +25%	Monitor every block; display ladder
Tier 1: Top-up	below 60%	about +25%	Draw AUSD reserve buffer to restore ratio to 100%; if empty, proportional deleverage; notify user
Tier 2: Deleverage	below 40%	about +43%	Forced proportional deleverage back to 100%; block new entries; push alert
Tier 3: Emergency unwind	below 25%	about +60%	Close the perp (reduce-only) and sell all spot, return to 100% AUSD; write EmergencyUnwind; lock the position

Tier	Trigger (margin ratio)	Approx. price rise at 1x	Action
Liquidation (what we prevent)	MMR, about 5% to 10%	about +82% to +90%	Exchange liquidation engine force-closes the short



Risk Guard tiers · 4 states, 2 recovery loops

Tiers 1 and 2 loop back to Normal once the ratio is restored; Tier 3 ends in a locked, all-AUSD position.

Thresholds scale with leverage. Tier 1, 2 and 3 sit at 60%, 40% and 25% of the initial margin ratio. At the 2x opt-in the ladder becomes 30%, 20% and 12.5%: Tier 3 fires near +33% while liquidation sits near +36% to +43%, leaving only a few points of room. That is why 2x is an advanced setting gated by the latency rule below.

Configuration	Notional / capital	Gross yield if funding is 15% APR	Price rise to liquidation (MMR 10% / 5%)	Role
1x with 10% AUSD reserve buffer	45%	6.75%	+82% / +90%	Default
1x, no buffer	50%	7.5%	+82% / +90%	Allowed
2x, no buffer	66.7%	10%	+36% / +43%	Advanced opt-in

Safety costs yield, and the app says so on the Start Earning screen.

Guard components

- **Price source:** margin ratio is computed from Perpl's mark price, since that is what its liquidation engine uses, with Kuru mid as a cross-check. Tiers 1 and 2 use a short TWAP to avoid wick-triggered churn; Tier 3 fires on two consecutive reads.
- **Pre-trade filter:** refuse entry on tokens whose realized volatility or 24-hour range exceeds a cap, with thin spot depth, with negative trailing funding, or while AUSD is off peg.
- **Latency (gap) rule:** the distance between the Tier 3 trigger and liquidation must exceed the largest price move expected in one guard cycle plus execution time (target under 2 seconds on Monad), times a safety factor. If a token is too volatile for that gap, the app lowers its maximum leverage automatically.
- **Emergency order of operations:** Tier 3 sends both legs in the same tick. Rule: never leave a naked short in a rising market. The perp close retries first with widening slippage up to a hard cap; the spot sale runs alongside but never completes alone. If a leg still fails, the guard escalates to a maximum-priority perp close and alerts the user.
- **Atomicity, stated honestly:** the Tier 3 trigger lives in the Risk Guard (app or keeper), not in the CarryLedger contract, because the ledger cannot call an API-matched venue. The unwind is near-simultaneous (both legs within one or two Monad blocks), not atomic. If Perpl and Kuru turn out to be callable from one onchain transaction, a single-transaction unwind contract becomes a stretch goal.
- **Exchange-resident backstop (verify support):** at entry, place a reduce-only stop on Perpl near the Tier 3 price and optionally a resting Kuru limit sell at the same price, so the exit still happens if the app is closed, offline or the key fails. Resting orders must be replaced after any deleverage. Treated as P2 until Perpl stop-order support is confirmed.
- **Always-on protection vs the Mera bounty:** Tiers 1 to 3 run in the app (the bounty-safe core), with push alerts, and the exchange-resident backstop covers app downtime. An opt-in remote keeper stays Phase 2.
- **Circuit breakers:** a price older than 5 seconds freezes entries but never blocks Unwind; negative funding for K intervals triggers a normal exit, not an emergency; AUSD deviation beyond its cap blocks entry and alerts.
- **Onchain proof:** each guard action writes `GuardTriggered(positionId, tier, marginRatioBps, action)` or `EmergencyUnwind(positionId, reason)` through CarryLedger, and Envio indexes them for the Position screen and the Activity log.
- **UI:** a margin ladder bar with the three tier lines and the current ratio, a plain distance-to-liquidation figure in percent, and an incident log.

- **Testing:** a stress harness replays price paths (+25%, +45% and +65% over 1, 5 and 30 seconds, plus a mean-reverting wick) through the margin math with simulated latency. Unit tests cover each tier transition, a failed leg, partial fills and an empty reserve buffer. A small-size drill forces Tier 1 by temporarily raising its trigger.

6. Strategy Engine Specification

6.1 Market eligibility

A market qualifies only if all hold:

1. Tradable on both Kuru (spot) and Perpl (perp). **Verify overlap first.**
2. Kuru book depth within 0.5% of mid exceeds a multiple of the intended notional.
3. Trailing average funding is positive and funding volatility is below a cap.
4. Perp open interest and spread are above minimum thresholds.

Hackathon scope: support one or two markets, hard-coded allowlist, with the eligibility filter running live to prove the logic.

6.2 Entry sequence and leg risk

The two legs cannot settle in one atomic transaction if the perp leg is API-matched, so the engine manages leg risk explicitly:

1. Pre-checks: balances, allowance, depth, funding, policy limits, plus the Risk Guard entry filter (volatility cap, AUSD peg, gap rule) and funding of the reserve buffer.
2. Perp leg first with an IOC limit order at a bounded price (the perp can be closed instantly with a reduce-only order, while the thinner spot book is harder to exit).
3. On confirmed fill, buy spot on Kuru up to the slippage cap, sized to the filled perp quantity.
4. If the spot leg fails or partially fills beyond a timeout (default 3 seconds), immediately reduce the perp to match, or close it.
5. Record the result via `openHedge`.

This ordering is a design choice; make it configurable and test both orders on small size. After entry, the Risk Guard arms its tiers and, where supported, the exchange-resident backstop.

6.3 Monitoring and rebalancing

Trigger	Condition (defaults)	Action
Hedge drift	Spot vs perp base quantity differ by more than 1%	Rebalance the smaller leg to match
Margin stress (Tier 1)	Margin ratio below 60% of its initial value	AUSD reserve first, else shrink both legs and add proceeds to margin
Deleverage (Tier 2)	Margin ratio below 40% of its initial value	Forced proportional deleverage to 100%, block new entries, alert
Emergency (Tier 3)	Margin ratio below 25% of its initial value, two consecutive reads	Close perp and sell all spot, return to AUSD, lock the position
Funding decay	Funding below floor for K consecutive intervals	Auto-unwind
Spread blowout	Spot/perp basis beyond cap	Pause new entries, alert
Stablecoin risk	AUSD price deviates beyond cap	Alert, block entry
Stale data	Funding or price feed older than N seconds	Block entry, keep Unwind available

The margin rows above are the entry points to the Risk Guard in section 5.5, which defines the full ladder. Key mechanism for the pitch: when price spikes, the short loses on the perp while the spot leg gains, but those gains are not automatically available as margin. The guard uses the AUSD reserve first, then shrinks both legs together so the position stays delta-neutral. Speed is the safety feature, which is the Monad argument in one sentence.

6.4 Exit

Unwind closes the perp (reduce-only), sells spot on Kuru, settles, writes `closeHedge`, and returns AUSD. The force unwind skips optimizations and accepts higher slippage; it is the same path the Tier 3 emergency unwind uses (section 5.5).

7. Data Layer (Envio)

Envio must drive a core feature, not decorate. Plan:

- Index CarryLedger events (HedgeOpened, HedgeRebalanced, HedgeClosed, PolicySet, GuardTriggered, EmergencyUnwind).
- Index Kuru Trade events for the user's fills to reconcile spot execution price.
- Expose GraphQL queries: positions by owner, rebalance history, realized PnL, hedge-drift time series, margin-ratio history and guard incidents.
- The Position and Activity screens read from Envio. The README should show the query used by each screen.

If the Perpl leg is API-only, reconcile its fills from the API and store an attested summary in the ledger event so history is still unified.

8. Bounty Mapping

Bounty	Prize	What it asks	How Carry satisfies it	Evidence to show
Agora: Best Mobile Trading App	$10,000	Mobile app, Mera auth, AUSD balance, trades via Perpl	Expo app, Mera login, AUSD-denominated home screen, perp leg on Perpl	Device screen recording, AUSD balance, Perpl fill
Kuru: Next Consumer Trading App	$5,000	Focused spot product routing through Kuru order book	Spot leg plus a standalone Spot tab on Kuru's book	Order book view, Kuru Trade tx hashes
Perpl: Best use of API	$5,000	Production-ready bot or automation	Strategy engine plus Risk Guard with retries, idempotency, rate-limit handling, stress tests, reconnects	Test suite and stress-harness output, logs of automated guard actions

Bounty	Prize	What it asks	How Carry satisfies it	Evidence to show
Monad: Best Mera-Powered UX	$2,500	Mera is the whole account layer	Passkey only, no custody backend in core build	Fresh-device passkey sign-in on video
Monad: One Passkey, Many Keys	$2,500	Creative non-wallet PRF use	PRF-encrypted Perpl API key vault	README section explaining the scheme
Envio: Best Use	$1,000	Envio drives a core feature	Positions, PnL, drift charts via HyperIndex	Queries in README
Nansen (stretch)	$5,000	Product that goes beyond raw data	Smart-money flow as a risk flag on candidate markets	Only if time remains

Combined ceiling for the core stack is $26,000, plus $5,000 if the Nansen stretch lands, plus the Track 01 prize ($10,000 per top-3 slot) and overall ($25,000). These are ceilings, not expectations. The Perpl 'Analytics / Risk Tool' bounty ($3,000, Track 01) is a possible add-on because the Position screen is effectively a risk dashboard; check its text before claiming it. Privy and Dynamic bounties are excluded because they conflict with the Mera account layer.

9. Judging Criteria Alignment

Criterion (20% each)	Plan	Honest risk
Product quality and completeness	Polished mobile flow, three real states (idle, active, unwinding), clear empty/error states	Two venue integrations in three days
Technical excellence	Small audited-style ledger, tests, clear leg-risk handling, tiered Risk Guard with stress harness, security notes	Perp leg off-chain dependence

Criterion (20% each)	Plan	Honest risk
Monad integration	Dual-venue sub-second execution, tight gas limits, Envio, rebalancing economics	Must show real onchain txs on video
Track fit	Financial primitive and algorithmic trading, packaged for non-traders	Strategy is known; novelty is packaging
Innovation and impact	Carry as a consumer primitive, passkey-secured automation, proof-of-hedge	Judges may ask about Hyperliquid-style vaults; answer: non-custodial, portable policy

Sponsor bounties weight requirements alignment at 40% and technical implementation at 30%, so every bounty row in section 8 needs a visible artifact in the demo and README.

10. Submission Compliance Checklist

- [ ] Public GitHub repo with a visible commit history inside Sept 1 to Oct 13. Commit early and often; the history is evidence. Use descriptive messages per feature (for example 'feat: Perpl IOC entry with timeout flatten') and never one giant commit.
- [ ] OSI-approved license (MIT recommended).
- [ ] README discloses any pre-existing code or libraries and every AI code-generation tool used.
- [ ] Contracts deployed on Monad testnet or mainnet, verified on the explorer, addresses and live tx hashes in the submission and README.
- [ ] 3-minute public video (Loom, YouTube or Vimeo) of the live product with Monad transactions visibly occurring. No slides, Figma or mockups.
- [ ] README includes architecture diagram, problem statement, tech stack, reproducible setup, `.env.example`, RPC settings and explorer links.
- [ ] Single submission, single track (Track 01). Bounty form fields completed per sponsor.
- [ ] Submit with at least 8 hours of buffer. Deadline is 11:59 PM ET on Oct 13 (06:59 AM Oct 14 in Istanbul) and is strictly enforced. Target: submit by 14:00 ET (21:00 Istanbul) on Oct 13.

11. Demo Video Script (under 180 seconds)

Time	Content
0:00 to 0:30	Problem: carry is powerful but needs two venues and constant babysitting. Introduce Selin. Show the one-line pitch.
0:30 to 1:20	Live on a physical phone: Face ID sign-in (Mera), AUSD balance, tap Start Earning, both legs execute, explorer shows Kuru and CarryLedger transactions.
1:20 to 2:00	Position screen: hedge ratio, margin health, real funding already accrued from a position opened earlier. Lower the Tier 1 trigger to just above the current margin ratio to force a real, automatic guard action on the live position, and show the margin ladder.
2:00 to 2:40	Architecture in one diagram: Mera, engine, Kuru, Perpl, ledger, Envio. Call out PRF-encrypted key, tight gas limits, sub-second legs.
2:40 to 3:00	Unwind in one tap, funds back in AUSD. Close with roadmap.

Important: funding accrues slowly, so open a real small position by October 11 so real funding payments exist before you record. Do not fake accrual. If you show a ticking earnings counter, label it 'estimated', compute it from the live funding rate, position size and elapsed time, and show realized funding payments separately. Say the design 'shrinks the one-leg window to about a second', never 'eliminates leg risk', because the perp leg may be API-matched.

12. Seventy-Two-Hour Build Plan

Assumes one expert developer working with AI tooling. Times are relative to starting now (Oct 10).

The cut list: built live, cut, never faked

Judges see four things: the video, the live transaction hashes, the README and the passkey flow. They will not run a six-month backtest, so every hour goes to those four, and nothing on the demo path is faked. What is not live is cut or clearly labeled.

Component	Build live (must have)	Cut or defer
Passkey and PRF	Mera sign-in on the target phone; PRF-encrypted Perpl key vault (section 5.4)	Multi-device sync, social recovery, guardians
Monad execution	Both legs executed from the app: Kuru spot buy and Perpl short, with real tx hashes saved in TX_LOG.md	A standalone off-chain keeper service; use an in-app 'Rebalance now' button as the guaranteed manual path
AUSD accounting	Real AUSD (mainnet with small amounts, or a testnet faucet if one exists)	Fiat ramps and bridges. A mock token is the last resort: label it MockAUSD, disclose it, and expect to lose the Agora AUSD requirement
Funding yield	Live Perpl funding rate driving the displayed APR; realized funding from a position opened early	Waiting for funding cycles during the video. An 'estimated accrual' counter is allowed only if labeled as an estimate (live rate x notional x elapsed time) and shown apart from realized funding
Contracts	CarryLedger: non-custodial, deployed, verified, emitting real events	A pooled CarryVault with deposit and withdraw: it adds custody and audit scope, and a mock vault is the simulation the rules penalize
Risk Guard	Margin ladder display, Tier 3 emergency unwind, manual Rebalance now; Tiers 1 and 2 if time allows	Historical backtests; publish only the small stress-harness results table in the README

Component	Build live (must have)	Cut or defer
UI	Dark, high-contrast fintech look in three screens: Unlock Vault, dashboard (AUSD balance, live APR, net delta, Kuru and Perpl leg cards), Position with Unwind	Onboarding carousels, deep settings, a web dashboard

Adopted from outside advice: log every tx hash from the first test; 'Unlock Vault' copy on the passkey screen; a net-delta tile computed from live leg values and highlighted when within 1% of zero; descriptive commits; the manual Rebalance now button; an ASCII or Mermaid architecture diagram in the README.

Rejected, with reasons:

- Simulating funding accrual as if it were real: it conflicts with the 'live product, no simulated mockups' rule and with the honesty standard in section 1.
- Mock AUSD as the default: the Agora bounties are about a real AUSD balance.
- A custody vault contract: more scope, more audit surface, no extra rubric points.
- Raw `navigator.credentials` with a dummy key in localStorage: it does not run in Expo and would not impress a security-minded judge.
- 'Parallel EVM eliminates leg risk': it shortens the one-leg window to about a second; it does not remove it.
- Submitting only 3 hours before the deadline: this plan targets 8 hours or more (14:00 ET on Oct 13).
- A web-only mobile viewport as the Agora entry: confirm the bounty text says 'mobile app' before relying on it; Expo is the safe route.

Hours 0 to 2: kill gates (stop and pivot if any fail)

1. A token is listed on both Kuru and Perpl with usable depth.
2. You can create a Perpl account and a trade-only API key, place a tiny order, and read mark price, margin ratio and the maintenance margin requirement from the API.
3. AUSD is available where you will deploy (confirm testnet vs mainnet; mainnet with small amounts may be safer for verifiable transactions).
4. Mera runs on your target phone (iOS 18+ or Android 9+ with a PRF-capable passkey provider). React Native needs a shared rpId with a web origin, so you need a domain and platform association files. Start this immediately; it is the most common time sink.

If gate 1 fails, switch to the cross-border payments project, which reuses the Mera and AUSD work.

Day 1 (Oct 10 to 11): foundation

- Repo, license, CI, Expo scaffold, Foundry project.
- Mera sign-in on device; AUSD balance read.
- CarryLedger contract plus tests; deploy and verify.
- Perpl client (auth, balance, place and cancel order, WebSocket). Kuru client (book, buy, sell).
- Open the first small real position before ending the day, to start accruing real funding. Log every tx hash and explorer link in TX_LOG.md as you go; the README and the video both need them.

Day 2 (Oct 11 to 12): engine and screens

- Entry sequence with leg-risk handling; unwind; rebalancer with drift triggers; Risk Guard tiers 1 to 3 with the AUSD reserve buffer.
- PRF-encrypted API key vault.
- Envio indexer and GraphQL queries; Position and Activity screens.
- Spot tab on Kuru.
- Error states and retry paths. Write unit tests and the stress harness for sizing, drift, margin math, tier transitions and failure sequences.

Day 3 (Oct 12 to 13): polish, proof, submit

- Morning: bug bash on a real device, tight gas limits, rate-limit handling, copy and visual polish.
- Midday: README (diagram, setup, addresses, bounty evidence section), disclosures, `.env.example`.
- Afternoon: record and edit the video; upload publicly.
- By 14:00 ET (21:00 Istanbul): submit, fill every bounty form field, verify all links open in a private window.
- Remaining buffer: fix only what the checklist flags. No new features.

Priority cut list

Priority	Items
P0 (must ship)	Mera login, AUSD balance, one market, entry, unwind, Tier 3 emergency unwind, manual Rebalance now button, margin ladder display, CarryLedger, Envio-backed Position screen, video, README
P1	Tier 1 and 2 guard (reserve buffer, deleverage), stress harness, PRF key vault, Spot tab, break-even calculator
P2	Second market, exchange-resident backstop, push notifications, Nansen risk flag, remote keeper

13. Risk Register

Risk	Likelihood	Impact	Mitigation
No market overlaps on Kuru and Perpl	Medium	Fatal	Gate 1 in the first two hours; pivot plan ready
Short leg liquidated in a price spike	Low to medium	Critical	1x default, AUSD reserve buffer, three-tier ladder, Tier 3 emergency unwind well above liquidation (section 5.5)
Spot top-up breaks the hedge (position turns net short)	Medium	High	Always shrink both legs together; unit test that delta stays within 1% after every guard action
App closed or offline during a spike	Medium	High	Exchange-resident backstop stops where supported; push alerts; opt-in remote keeper in Phase 2

Risk	Likelihood	Impact	Mitigation
Emergency unwind half-fills (one leg fails)	Low	High	Perp close first with a widening slippage cap, never leave a naked short in a rising market, alert the user
Guard false trigger from a price wick	Medium	Medium	Mark-price TWAP for Tiers 1 and 2, two consecutive reads for Tier 3, cooldown after each action
Wrong maintenance margin or liquidation assumption	Medium	High	Read the real values from Perpl on day 1; ladder thresholds are ratios relative to them
Perp API access or limits block automation	Medium	High	Early key creation; backoff and idempotent order IDs
Leg fails, naked exposure	Medium	High	Perp-first ordering, 3-second timeout, auto-flatten, tests
Funding negative during demo	Medium	Medium	Auto-exit logic shown; frame as feature; do not promise APR
Mera/PRF fails on a device	Medium	High	Test on target phone in hour 1; keep a second device
Thin spot liquidity causes slippage	High	Medium	Small sizes, limit orders, depth filter
Scope creep	High	High	Enforce P0 list; no new features after Day 2
Funding or price feed stale	Low	Medium	Staleness guard blocks entry, keeps Unwind
Smart-contract bug	Low	Low	Ledger holds no funds; tests; custom errors

14. Security and Quality Notes

- Ledger is non-custodial by design; document that clearly.

- Treat the Perpl key as a secret: trade-only, revocable, encrypted at rest under PRF, never logged.
- Validate every external number (prices, funding, balances) before acting; reject NaN, zero and stale values.
- Idempotent order placement with client-generated IDs so retries cannot double-open.
- Test matrix: sizing math, partial fills, one-leg failure, timeout, each guard tier, hedge-delta check after every guard action, unwind with stale data, duplicate retry.
- Use small real amounts; display a clear 'hackathon prototype, not financial advice' notice in app and README. Add a Risk Guard fire drill: force each tier on small size before recording the video.

15. Open Questions and Unverified Assumptions

1. Exact market overlap between Kuru and Perpl, and funding interval on Perpl.
2. Whether AUSD is deployed on Monad testnet, and whether Perpl accepts it as collateral. Fallback order: real AUSD on mainnet with small amounts, then a testnet faucet, and only last a clearly labeled mock token, which forfeits the AUSD requirement of the Agora bounties and must be disclosed.
3. Whether Perpl matching and settlement are fully onchain or API-matched (decides whether the Tier 3 unwind can be atomic or only near-simultaneous, and what the pitch may claim).
4. Whether a TypeScript Kuru SDK exists; the SDK found is Python, so direct contract calls via viem may be needed.
5. Whether the Mera bounty permits an opt-in remote keeper.
6. How many winners each sponsor bounty has (the listing does not say).
7. Whether the Perpl Analytics / Risk Tool bounty can be claimed alongside the API bounty.

16. Roadmap After the Hackathon

The top teams are invited to a residency, so frame a credible next step:

- Pooled, audited vault so users do not run their own legs.
- More markets and tokenized-asset carry as listings arrive.
- Remote keeper with scoped, expiring permissions, and a single-transaction emergency unwind if both venues become callable from one contract.

- Cross-venue funding comparison and multi-leg basis trades.
- Insurance-style reserve and formal audits before any public mainnet launch.

17. One-Paragraph Pitch (for the submission form)

Carry turns a professional two-venue strategy into a one-tap consumer product on Monad. Users sign in with a passkey, hold AUSD, and earn market-neutral funding yield by pairing a Kuru spot position with a Perpl short, while an automated engine keeps the hedge balanced, defends the short leg with a three-tier margin guard and an emergency exit, and leaves when conditions change. Every hedge action is recorded onchain and indexed with Envio, the Perpl key is encrypted under the user's passkey, and Monad's sub-second execution is what makes continuous rebalancing safe from a phone.